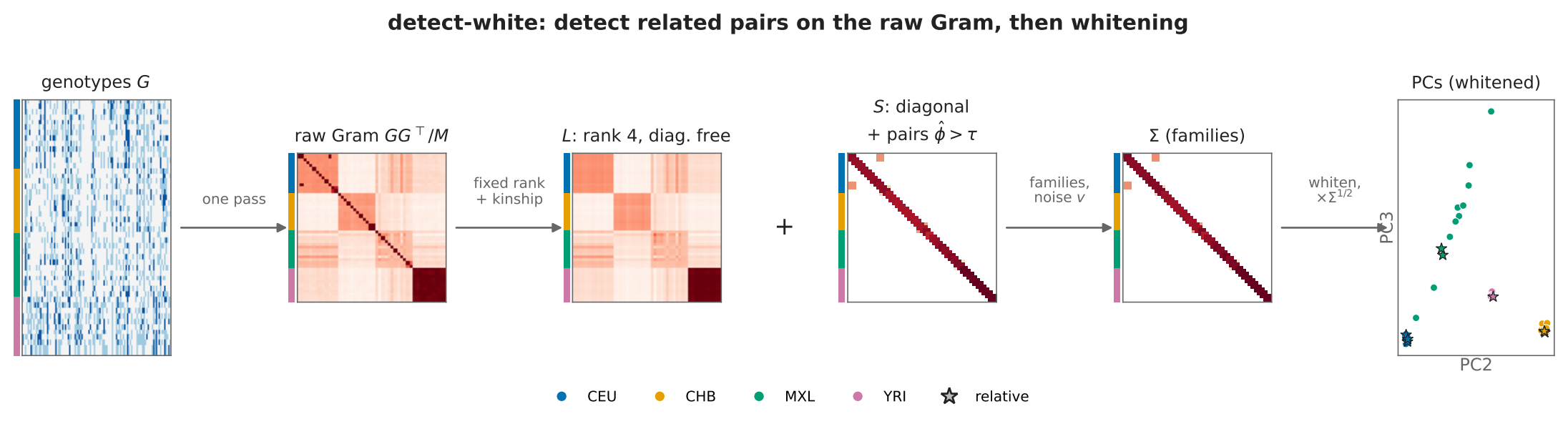
detect-white: detect related pairs on the raw Gram, then whitening
genotypes G
one pass
raw Gram $GG^\top/M$
fixed rank
+ kinship
L: rank 4, diag. free
+
S: diagonal
+ pairs $\hat{\phi} > \tau$
families,
noise v
Σ (families)
whiten,
$\times\Sigma^{1/2}$
PCs (whitened)
PC3
PC2
CEU
CHB
MXL
YRI
relative